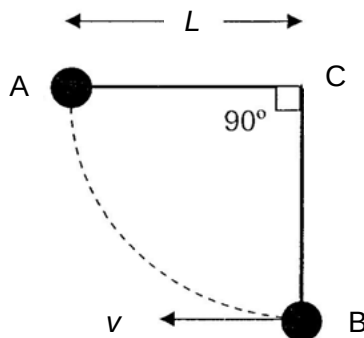


- 9 A simple pendulum consists of a bob of mass m at the end of a light and inextensible thread of length L . The other end of the thread is fixed at C. The bob swings through point B with velocity v and just reaches A, where the string is just taut.



What is the tension in the thread as the bob passes point B?

- A mg B $2mg$ C $3mg$ D $4mg$

Ans: C

gain in GPE = loss in KE

$$mgL = \frac{1}{2}m(v^2 - v_A^2) \quad [v_A = 0]$$

$$mgL = \frac{1}{2}mv^2$$

$$mv^2 = 2mgL$$

$$T - mg = \frac{mv^2}{L} = \frac{2mgL}{L}$$

$$T = 3mg$$

- 10 An object X has weight W on the surface of the Earth. How much energy is needed to raise X